SMITH ENGINEERING
Queen's University

Mechatronics and Robotics Engineering

Calculation Sheet

Project: SAMI
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Area: LiDAR readings
Subject: MREN 403

No.
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Checked by: Toby Copeland
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Sheet 1 of 1

Range = 270°
Frequency = 50 Hz ⟶ updates every 0.2seconds
Angle resolution = 0.5°

270/0.5 +1 = 541 data points
(+1 account for starting and ending measurements)

x = distance $(\cos(\theta))$
y = distance $(\sin(\theta))$
} polar to cartesian coordinates

range/2 = 270 ÷ 2 = 135° → starts at $-135° = \theta_5$

if we want θ of data point 4:

$\theta_4 = \theta_5 + 0.5 \times 3$

$\theta_4 = -135 + 1.5 = -133.5°$

Then repeat for all the angles.